Glomerulopathy With Fibronectin Deposits 2 (GFND2): Comprehensive Disease Characterization Report

Disease: Glomerulopathy With Fibronectin Deposits 2 (GFND2) **MONDO ID:** MONDO:0011165 | **OMIM:** 601894 | **Causal gene: FN1 (fibronectin 1, 2q35; HGNC:3778) Category:** Mendelian (autosomal dominant)

Summary

Glomerulopathy with fibronectin deposits 2 (GFND2; OMIM 601894, MONDO:0011165) is a rare autosomal dominant hereditary glomerular disease caused by heterozygous variants in **FN1**, the gene encoding **fibronectin 1**, located at chromosome 2q35. The disease is defined histopathologically by the pathognomonic feature of massive glomerular deposits of fibronectin in the mesangium and subendothelial space, producing a membranoproliferative glomerulonephritis (MPGN) type III-like architecture. Clinically, patients present with proteinuria (frequently in the nephrotic range), microscopic hematuria, hypertension, and slow but relentless progression to end-stage renal disease (ESRD), with terminal renal failure occurring at a median age of approximately 47 years in a large kindred.

Across the largest aggregated series of 72 genetically confirmed patients, the **fibronectin heparin-binding domain was identified as the mutational hotspot** [PMID: 42068364]. The recurrent, canonical variant is **NM_212482.4 c.2918A>G (p.Tyr973Cys / Y973C)**, reported across multiple unrelated patients and populations. Functional studies indicate a **gain/alteration-of-function mechanism**: at least one variant (c.3415G>A) increases fibronectin binding to integrin while decreasing binding to collagen type IV, driving glomerular-specific fibronectin deposition [PMID: 36774238]. A key pathogenic insight is that GFND2 **recurs in kidney allografts**, implicating circulating (plasma-derived) mutant fibronectin as the deposit source rather than a purely local kidney defect [PMID: 35774511, 22747478].

There is no disease-specific or curative therapy. Management is supportive nephroprotection—RAAS blockade (ACE inhibitors/ARBs), the angiotensin receptor-neprilysin inhibitor sacubitril/valsartan, and SGLT2 inhibitors as adjuncts—with inconsistent responses to immunosuppression. ESRD is managed by dialysis and kidney transplantation, though allograft recurrence is well documented. Research models are limited to patient-derived induced pluripotent stem cells (iPSCs); no established GFND2-specific animal model exists, partly because constitutive Fn1 knockout is embryonic lethal in mice. The disease is frequently misdiagnosed as MPGN type I/III, thrombotic microangiopathy (TMA), lupus, or IgA nephritis until fibronectin immunohistochemistry and FN1 genetic testing establish the correct diagnosis.

> **Evidence note:** GFND2 is an ultra-rare disease known from <100 genetically confirmed patients worldwide. Most evidence is Level IV/V (case reports, small families, mechanistic in vitro work). Quantitative epidemiology, controlled trials, and formal penetrance estimates are **not available**.

Key Findings

Finding 1 — GFND2 is an autosomal dominant glomerulopathy caused by FN1 mutations

GFND2 is a Mendelian, autosomal dominant kidney disease caused by heterozygous pathogenic variants in **FN1 (fibronectin 1)**, located at chromosome **2q35**. Multiple independent human cohorts and case reports have linked heterozygous FN1 variants to glomerulopathy with fibronectin deposits (GFND2, OMIM 601894). A comprehensive literature review of **72 genetically confirmed patients** established that the **fibronectin heparin-binding domain is a mutational hotspot** [PMID: 42068364]:

> "The comprehensive literature review of 72 patients revealed that the heparin-binding domain is a mutational hotspot" [PMID: 42068364]

The classic recurrent variant is **NM_212482.4 c.2918A>G (p.Tyr973Cys / Y973C)**, reported across multiple unrelated patients and distinct populations [PMID: 41642109, 39285372, 42675453]:

> "the FN1 gene was heterozygous for NM_212482.4 (c.2918A>G), that is, Y973C mutation" [PMID: 41642109]

This establishes both the gene–disease relationship and the recurrent hotspot that anchors molecular diagnosis. **Suggested ontology terms:** HGNC:3778 (FN1); MONDO:0011165; UBERON:0002113 (kidney).

Finding 2 — Clinical phenotype: proteinuria, hematuria, hypertension, slow progression to ESRD (MPGN-like)

The clinical syndrome is consistent across case series: the main manifestations are **proteinuria** (often nephrotic-range), **microscopic hematuria**, **hypertension**, and **progressive decline in renal function** culminating in **end-stage renal disease** [PMID: 41656823]:

> "Proteinuria, hematuria, hypertension and progressive deterioration of renal function are the main clinical manifestations, which eventually develop into end-stage renal disease (ESRD)" [PMID: 41656823]

Histology shows an **MPGN type III-like pattern** with massive glomerular fibronectin deposits. The natural history is one of slow progression: in a large kindred, the **median age to terminal renal failure was ~47 years** [PMID: 10964510]:

> "an autosomal dominant kidney disease that leads to terminal renal failure at a median age of 47 years" [PMID: 10964510]

Onset is typically in the 2nd–5th decade (ages ~10–50 years) but the reported range is wide, from age 4 to 75 [PMID: 38254040, 39859354, 39285372]. **Suggested HPO terms:** HP:0000093 (Proteinuria), HP:0000790 (Hematuria), HP:0000822 (Hypertension), HP:0000083 (Renal insufficiency), HP:0003774 (Stage 5 chronic kidney disease), HP:0000100 (Nephrotic syndrome).

Finding 3 — Mechanism: gain/alteration of matrix-binding function causing glomerular fibronectin deposition

Functional characterization supports a **gain-of-function** / **altered-binding** mechanism rather than simple loss of function. The FN1 variant c.3415G>A was shown to cause glomerular fibronectin-specific deposition in a gain-of-function manner, **increasing** fibronectin binding to integrin (maintaining podocyte adhesion) while **decreasing** binding to collagen type IV (COL4A3/4) [PMID: 36774238]:

> "a novel mutation (c.3415G>A) in FN1 that causes glomerular fibronectin-specific deposition in a gain-of-function manner" [PMID: 36774238]
>
> "this variant increased fibronectin's ability to bind to integrin, thereby maintaining podocyte adhesion" [PMID: 36774238]

The disease-defining histology is **massive mesangial and subendothelial fibronectin deposits** that are **PAS-positive, Jones silver-negative, and Congo red-negative**, with MPGN type III-like architecture [PMID: 10964510, 40529320]. Systemic (extra-renal) fibronectin deposition has been demonstrated in gastric mucosal extracellular matrix, indicating the defect is not confined to the kidney [PMID: 42068364]. **Suggested GO terms:** GO:0005178 (integrin binding), GO:0005518 (collagen binding), GO:0007160 (cell-matrix adhesion), GO:0030198 (extracellular matrix organization). **Suggested CL terms:** CL:0000653 (podocyte), CL:1000742 (glomerular mesangial cell). **Suggested UBERON:** UBERON:0001229 (renal glomerulus).

Finding 4 — Allograft recurrence implicates circulating (plasma) fibronectin

A pivotal pathophysiological insight is that GFND2 **recurs in kidney allografts**, strongly implicating **circulating/plasma-derived mutant fibronectin** in the pathogenesis rather than a kidney-intrinsic defect alone. Multiple post-transplant recurrence cases are documented, including the first Japanese case [PMID: 22747478] and subsequent reports [PMID: 35774511, 37852577, 38053039], with recurrence occurring from weeks–months to ~27 months post-transplant:

> "Given the persistence of abnormal circulating fibronectin levels, FNG can relapse following renal transplantation. The circulating fibronectin deposits on grafts, and renal function progressively deteriorates after recurrence" [PMID: 35774511]

In one case, the mutant FN1 missense variant **reduced** fibronectin binding to heparin, endothelial cells, and podocytes and **impaired stress-fiber formation**, with persistent abnormal circulating fibronectin depositing on the graft [PMID: 35774511]:

> "a missense mutation in the fibronectin 1 (FN1) gene caused reduced binding to heparin, endothelial cells, and podocytes and impaired stress fiber formation" [PMID: 35774511]

Native disease typically progresses to kidney failure over 15–20 years. **This finding is clinically important:** because the pathogenic protein circulates, transplantation does not cure the underlying disease, and graft recurrence must be anticipated. **Suggested GO terms:** GO:0008201 (heparin binding), GO:0032956 (regulation of actin cytoskeleton organization).

Finding 5 — Variant spectrum: dominant missense hotspots plus splice/in-frame deletions; frequent misdiagnosis

By 2025, approximately **ten exonic and one intronic pathogenic FN1 variants** had been catalogued, but **genotype–phenotype correlation data remain lacking** [PMID: 40120032]:

> "Currently, ten exonic and one intronic pathogenic variants in the fibronectin 1 gene have been identified; however, genotype-phenotype correlation data are lacking" [PMID: 40120032]

Variant types include: - **Recurrent missense:** Y973C (c.2918A>G); W1925R; G417V [PMID: 39859354] - **In-frame deletion:** c.5749_5751delACT (p.Thr1917del), de novo [PMID: 42068364] - **Splice-site variants:** intron 36 c.5888-2A>G (familial) [PMID: 40120032]; c.5888-1G>C [PMID: 37801747]

De novo occurrence is documented [PMID: 38254040, 42068364]. Additional reported features include **hyperkalemic distal renal tubular acidosis** [PMID: 41642109] and **marked hyperlipidemia with intracapillary lipid deposition** [PMID: 42967636]. The disease is **frequently misdiagnosed** as MPGN type I/III, thrombotic microangiopathy, lupus nephritis, or

IgA nephritis before fibronectin immunohistochemistry and FN1 genetics clarify it [PMID: 41642109, 35774511, 37852577]. Pathologic-genomic correlation—including glomerular proteomics detecting mutant-allele fibronectin peptides—has been used to reclassify variants of unknown significance (VUS) [PMID: 37852577]:

> "genetic studies showed a variant of unknown significance in the fibronectin 1 gene. Proteomic analyses of the glomeruli ... demonstrated large amount of fibronectin with abundant accumulation of the peptide synthesized by the detected variant" [PMID: 37852577]

Finding 6 — Management is supportive/nephroprotective; no disease-specific therapy

No targeted or curative therapy exists. Management centers on **RAAS blockade** (ACE inhibitors/ARBs) to lower proteinuria and preserve function; in one elderly patient, conservative ARB therapy preserved function and reduced proteinuria over 2 years [PMID: 39285372]. The angiotensin receptor-neprilysin inhibitor **sacubitril/valsartan** has been used [PMID: 41642109; P 42967636], where it was combined with simvastatin with stable renal function and lipids over 17 months]. **SGLT2 inhibitors** are employed as adjuncts [PMID: 42675453]:

> "prednisolone and cyclosporine A were initiated for management of nephrotic syndrome. The patient subsequently achieved partial remission" [PMID: 42675453]
>
> "a sodium-glucose cotransporter 2 (SGLT2) inhibitor was added as adjunct therapy" [PMID: 42675453]

Immunosuppression (glucocorticoids ± cyclosporine, tacrolimus, mizoribine, azathioprine) gives **variable/incomplete responses**: partial remission with prednisolone + cyclosporine + SGLT2i [PMID: 42675453], incomplete remission with glucocorticoid + mizoribine + ARB [PMID: 40120032], and no clear response to RASi + steroids + tacrolimus in another case [PMID: 40529320]. End-stage disease is managed by dialysis and kidney transplantation, but FNG can

recur in the allograft [PMID: 22747478, 35774511]. **Suggested NCIT terms:** NCIT:C61602 (ACE Inhibitor), NCIT:C98079 (Angiotensin Receptor Blocker), NCIT:C1551 (glucocorticoid), NCIT:C2367 (Kidney Transplantation).

Finding 7 — Research models limited to patient-derived iPSCs; systemic deposition documented

A patient-derived **induced pluripotent stem cell line (XJHi001-A)** was generated from an FNG patient carrying heterozygous FN1 c.5888-1G>C to study pathogenesis and test therapies [PMID: 37801747]:

> "we generated an induced pluripotent stem cell (iPSC) line from peripheral blood mononuclear cells (PBMCs) of a 32-years-old female patient who carried a heterozygous NM_212482.2(C.5888-1G > C) mutation in the FN1 gene" [PMID: 37801747]

Otherwise, variant effects have been studied **in vitro** via binding assays (integrin up, COL4 down [PMID: 36774238]; heparin/endothelial/podocyte binding down, impaired stress fibers [PMID: 35774511]) and by **glomerular proteomics** [PMID: 37852577]. Systemic involvement is shown by fibronectin deposition in gastric mucosal extracellular matrix [PMID: 42068364]:

> "Gastric mucosal immunohistochemistry revealed specific fibronectin deposition within the extracellular matrix, indicating systemic involvement" [PMID: 42068364]

No GFND2-specific knockout/knock-in mouse disease model is reported in this literature set; notably, constitutive Fn1 knockout is embryonic lethal in mice, limiting simple loss-of-function modeling.

Finding 8 — Disease identity, locus heterogeneity, and diagnostic gold standard

GFND (MIM 601894) is an autosomal dominant kidney disease representing a **distinct entity of MPGN type III** characterized by the unique feature of massive glomerular fibronectin deposits [PMID: 10964510]:

> "It represents a distinct entity of membranoproliferative glomerulonephritis (MPGN) type III and is characterized by the unique feature of massive glomerular deposits of fibronectin" [PMID: 10964510]

An earlier GFND locus mapped to **chromosome 1q32 (GFND1)** in a large kindred with no causal gene identified [PMID: 10964510]; the gene-defined form is **FN1 at 2q35** = **GFND2**. This represents **locus heterogeneity** within the GFND phenotype. Definitive diagnosis rests on biopsy showing **PAS-positive, silver-negative, Congo red-negative** mesangial/subendothelial deposits that **stain strongly for fibronectin by immunohistochemistry (IST-4 positive / IST-9 negative pattern)**, with confirmatory FN1 sequencing and, in ambiguous cases, glomerular mass-spectrometry proteomics detecting mutant-allele fibronectin peptides [PMID: 40529320, 22747478, 37852577]:

> "extensive periodic acid Schiff-positive but Jones' methenamine silver-negative and Congo red-negative mesangial and capillary wall deposits, which stained strongly for fibronectin on immunohistochemistry" [PMID: 40529320]

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous germline (occasionally de novo) variant in FN1** (hotspot in the heparin-binding domain; e.g., Y973C / c.2918A>G) **leads to** production of a structurally altered fibronectin protein.
2. The mutant fibronectin **alters its matrix-binding profile** — demonstrated for c.3415G>A as *increased* integrin binding and *decreased* collagen type IV binding [PMID: 36774238], and for another variant as *decreased* heparin/endothelial/podocyte binding with impaired stress-fiber formation [PMID: 35774511]. (Mechanism is variant-specific; a unifying biochemical theme is dysregulated ECM assembly rather than loss of fibronectin per se — *inferred*.)

3. Altered binding **results in** failure of normal fibronectin incorporation into the glomerular extracellular matrix and **aberrant accumulation/deposition** of fibronectin in the mesangium and subendothelial space.
4. Because mutant fibronectin **circulates in plasma** (*inferred* from allograft recurrence), deposition is continuously fed from the circulation — **this is why transplantation does not cure the disease and grafts recur** [PMID: 35774511, 22747478].
5. Massive fibronectin deposits **lead to** expansion of the mesangium, thickening/duplication of capillary walls, and an **MPGN type III-like architecture** that distorts the glomerular filtration barrier.
6. Disruption of the filtration barrier **results in** proteinuria (often nephrotic-range) and hematuria; glomerular/vascular involvement **contributes to** hypertension.
7. Progressive glomerular injury and secondary tubulointerstitial fibrosis **lead to** slow decline in GFR and, over ~15–20 years (median terminal failure ~47 years), **end-stage renal disease** [PMID: 10964510, 35774511].
8. **Branch — systemic involvement:** the same circulating mutant fibronectin **deposits in extra-renal extracellular matrix** (e.g., gastric mucosa), indicating GFND2 is a systemic ECM disorder with a predominantly renal phenotype [PMID: 42068364].

Upstream vs downstream

Tier	Event	Evidence type
Upstream (initiating)	FN1 heterozygous variant (hotspot: heparin-binding domain)	Human genetics [PMID: 42068364, 41642109]
Upstream (molecular)	Altered fibronectin matrix-binding (integrin ↑/COL4 ↓; heparin ↓)	In vitro functional [PMID: 36774238, 35774511]
Intermediate	Circulating mutant fibronectin → glomerular deposition	Human (allograft recurrence) [PMID: 35774511, 22747478]
Downstream (tissue)	MPGN III-like remodeling, filtration-barrier disruption	Histopathology [PMID: 10964510, 40529320]
Downstream (clinical)	Proteinuria, hematuria, hypertension → ESRD	Human clinical [PMID: 41656823, 10964510]

Cell types and compartments

- **Cell types:** podocytes (CL:0000653), glomerular endothelial cells (CL:1001005), glomerular mesangial cells (CL:1000742); hepatocytes/plasma as the likely source of circulating fibronectin.
- **Anatomical sites (UBERON):** renal glomerulus (UBERON:0001229), glomerular basement membrane (UBERON:0002188), mesangium (UBERON:0005750), kidney (UBERON:0002113); extra-renal: stomach/gastric mucosa (UBERON:0001199).
- **Subcellular (GO CC):** extracellular matrix (GO:0031012), extracellular region (GO:0005576), actin cytoskeleton/stress fibers (GO:0001725).
- **Biological processes (GO BP):** extracellular matrix organization (GO:0030198), cell-matrix adhesion (GO:0007160), integrin-mediated signaling (GO:0007229).

Evidence Base

PMID	Title (abbrev.)	Contribution	Evidence type
10964510	Molecular cloning of critical region for GFND	Defines GFND as distinct MPGN III entity; 1q32 (GFND1) locus; median terminal failure ~47 y	Human genetics/ clinical
36774238	High-impact FN1 variant & decreased COL4 binding	Gain-of-function mechanism; integrin ↑/COL4 ↓	In vitro functional
35774511	Recurrent deposition in renal allografts	Circulating fibronectin drives recurrence; heparin/podocyte binding ↓	Human + in vitro
22747478	Recurrent FNG in renal transplant (first Japan)	Allograft recurrence; IST4+/IST9− diagnostic signature	Human clinical
37852577	Pathologic-genomic correlation	VUS reclassification via glomerular proteomics	Human clinical/ proteomics
42068364	Pediatric FNG, p.Thr1917del, systemic involvement	72-patient review: heparin-binding hotspot; gastric ECM deposition	Human review/clinical
40120032	Familial FNG, splice-site variant	Catalogs ~10 exonic + 1 intronic variants; no genotype-phenotype data	Human genetics/ clinical
41656823	Glomerulopathy with fibronectin deposits	Canonical clinical quartet → ESRD	Human clinical
41642109	FNG case + literature review	Documents recurrent Y973C; hyperkalemic distal RTA	Human clinical
39285372	FNG in elderly patient	Y973C; ARB preserved function over 2 y	Human clinical
42675453	Late-onset NS, 17-y course	Immunosuppression partial remission; SGLT2i adjunct	Human clinical
42967636	FNG with intracapillary lipid deposition	Hyperlipidemia; sacubitril/valsartan + simvastatin	Human clinical

PMID	Title (abbrev.)	Contribution	Evidence type
37801747	iPSC line XJHi001-A	Patient-derived disease model; c.5888-1G>C	In vitro model
38254040	FNG with MPGN features	De novo variant; IST-4 confirmation	Human clinical
39859354	FNG without typical features, G417V	Atypical/early presentation (4-y-old)	Human clinical
40529320	First African case report	Diagnostic histochemical signature	Human clinical
38053039	FNG in allograft biopsy	Allograft recurrence; prior MPGN misdiagnosis	Human clinical
42150947	FNG + chronic active ABMR in allograft	Allograft complexity; treatment-refractory	Human clinical

Overall evidence quality: The evidence base is composed almost entirely of **case reports, small case series, and one 72-patient aggregated literature review**, reflecting the rarity of the disease. There are no randomized trials, no prospective cohorts, and no formal prevalence/incidence estimates. Mechanistic evidence rests on a handful of in vitro binding studies and one patient-derived iPSC line. This evidence tier is appropriate for an ultra-rare Mendelian disorder but limits the strength of quantitative and therapeutic conclusions.

Section-by-Section Synthesis (Template Coverage)

1. Disease Information

GFND2 is a rare autosomal dominant hereditary glomerulopathy defined by massive glomerular fibronectin deposits. **Identifiers:** OMIM 601894; MONDO:0011165; causal gene FN1 (HGNC:3778, 2q35). **Synonyms:** glomerulopathy with fibronectin deposits type 2; FN1 fibronectin glomerulopathy; fibronectin glomerulopathy (FNG/FG). Information is derived from **aggregated disease-level resources and individual published patient case reports**, not EHR mining.

2. Etiology

Primary cause is genetic: heterozygous pathogenic FN1 variants (autosomal dominant). No environmental, infectious, or lifestyle cause is established. **Genetic risk:** causal FN1 variants (hotspot = heparin-binding domain; recurrent Y973C). No susceptibility loci, modifier genes, protective variants, or gene-environment interactions are documented for this ultra-rare monogenic disease. Note **locus heterogeneity**: a gene-unidentified GFND1 locus maps to 1q32 [PMID: 10964510].

3. Phenotypes

Phenotype	Type	HPO	Onset/severity/frequency
Proteinuria (often nephrotic)	Lab/ clinical	HP:0000093	Presenting sign; frequent/ near-universal
Hematuria (microscopic)	Lab/ clinical	HP:0000790	Common
Hypertension	Clinical sign	HP:0000822	Common
Progressive renal failure → ESRD	Clinical	HP:0000083 / HP:0003774	Progressive; median terminal ~47 y
Nephrotic syndrome/edema	Clinical	HP:0000100	Subset
Hyperlipidemia (reported)	Lab	HP:0003077	Case-level [PMID: 42967636]
Distal renal tubular acidosis (hyperkalemic)	Lab	HP:0008341	Case-level [PMID: 41642109]

Onset ranges from age 4 to 75, typically 2nd–5th decade; progression is slow and chronic. Quality-of-life data are not formally reported, but the trajectory toward dialysis/transplant implies substantial long-term morbidity.

4. Genetic/Molecular Information

Causal gene: FN1 (disease OMIM 601894; gene OMIM 135600). **Variant types:** missense (Y973C/c.2918A>G, W1925R, G417V), in-frame deletion (p.Thr1917del), splice-site (c.5888-2A>G, c.5888-1G>C). ~10 exonic + 1 intronic pathogenic variants catalogued [PMID: 40120032]. **Origin:** germline (autosomal dominant), with de novo occurrences [PMID: 38254040, 42068364].

Functional consequence: gain/alteration-of-function (altered matrix binding) [PMID: 36774238], not classic loss of function. No modifier genes, epigenetic mechanisms, or chromosomal abnormalities are established. Population allele frequencies are consistent with a very rare pathogenic variant set.

5. Environmental Information

No environmental factors, lifestyle factors, or infectious agents are implicated. GFND2 is a monogenic disorder; environmental contribution is **not applicable**.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** section above for the full ordered causal chain. Key pathway nodes: integrin binding (GO:0005178), collagen binding (GO:0005518), heparin binding (GO:0008201), ECM organization (GO:0030198). Upstream = FN1 variant + altered binding; downstream = glomerular deposition → filtration-barrier failure → ESRD. Circulating fibronectin is the inferred vehicle for continuous deposition.

7. Anatomical Structures Affected

Primary organ: kidney (UBERON:0002113), specifically the glomerulus (UBERON:0001229) — mesangium (UBERON:0005750) and subendothelial/capillary wall. **Cells:** podocytes (CL:0000653), mesangial cells (CL:1000742), glomerular endothelial cells. **Subcellular:** extracellular matrix (GO:0031012). **Extra-renal/systemic:** gastric mucosal ECM (UBERON:0001199) [PMID: 42068364]. Involvement is **bilateral** (systemic genetic disease).

8. Temporal Development

Onset: typically adult (2nd–5th decade); range age 4–75; **insidious/chronic**. **Progression:** slow and progressive to end-stage over ~15–20 years; median terminal renal failure ~47 years [PMID: 10964510, 35774511]. **Course:** chronic, progressive, lifelong; no spontaneous remission reported. Partial, treatment-associated proteinuria reduction is possible with nephroprotection/immunosuppression.

9. Inheritance and Population

Inheritance: autosomal dominant. **Penetrance:** appears high but not formally quantified; variable expressivity and wide age-of-onset range noted. De novo variants occur. **Epidemiology:** no formal prevalence/incidence estimates exist — an ultra-rare disease with

cases reported across Asian (Japanese, Korean, Chinese), European, and now African [PMID: 40529320] populations. No established founder effect, consanguinity role (AD disease), or sex predilection is documented. Genotype–phenotype correlation data are lacking [PMID: 40120032].

10. Diagnostics

Gold standard: kidney biopsy showing PAS-positive, Jones silver-negative, Congo red-negative mesangial/subendothelial deposits that stain strongly for **fibronectin by IHC (IST-4 positive / IST-9 negative)** [PMID: 40529320, 22747478], confirmed by **FN1 sequencing** (single-gene or panel/WES). Electron microscopy shows fibrillary/granular dense deposits. **Glomerular mass-spectrometry proteomics** can detect mutant-allele fibronectin peptides and reclassify VUS [PMID: 37852577]. **Differential diagnosis:** MPGN type I/III, thrombotic microangiopathy, lupus nephritis, IgA nephropathy, amyloidosis (excluded by Congo red negativity) — frequent misdiagnosis before fibronectin IHC/genetics [PMID: 41642109, 35774511, 37852577]. **Genetic testing** (single-gene FN1, renal gene panels, WES) is diagnostic and enables cascade family screening/counseling.

11. Outcome/Prognosis

Prognosis is **poor over the long term**: progressive decline to ESRD requiring dialysis/transplantation, median terminal renal failure ~47 years [PMID: 10964510]. No formal survival statistics beyond renal outcomes. **Prognostic factors** are not formally established (no genotype-phenotype correlation [PMID: 40120032]). Key complication is **allograft recurrence** after transplantation [PMID: 22747478, 35774511, 38053039], and concomitant antibody-mediated rejection can further threaten graft survival [PMID: 42150947].

12. Treatment

No disease-specific/curative therapy. **Supportive nephroprotection:** RAAS blockade (ACEi/ARB; NCIT:C61602/C98079) [PMID: 39285372]; sacubitril/valsartan [PMID: 41642109, 42967636]; SGLT2 inhibitors as adjuncts [PMID: 42675453]; statins for hyperlipidemia [PMID: 42967636]. **Immunosuppression** (glucocorticoids, cyclosporine, tacrolimus, mizoribine) yields variable/incomplete responses [PMID: 42675453, 40120032, 40529320]. **Renal replacement:** dialysis and kidney transplantation (NCIT:C2367) for ESRD, with recurrence risk. No approved gene, cell, or RNA-based therapy exists.

13. Prevention

No primary prevention (monogenic disease). **Secondary/tertiary prevention:** early diagnosis via biopsy + genetics, blood-pressure and proteinuria control to slow progression. **Genetic counseling** is central: autosomal dominant 50% transmission risk; cascade screening of relatives (e.g., retrospective recognition of affected deceased relatives previously labeled "type 1 MPGN" [PMID: 38053039]). Prenatal/preimplantation testing is theoretically possible once a familial variant is known.

14. Other Species / Natural Disease

FN1 is highly evolutionarily conserved (orthologs in mouse *Fn1*, etc.). No naturally occurring GFND2-equivalent disease in companion animals or wildlife is documented in this literature set. Veterinary/zoonotic relevance: **not applicable**.

15. Model Organisms

The principal model is a **patient-derived iPSC line (XJHi001-A)** carrying FN1 c.5888-1G>C [PMID: 37801747]. Mechanism is otherwise studied via **in vitro binding assays** and **glomerular proteomics**. **No GFND2-specific knockout/knock-in mouse model** is reported; constitutive *Fn1* knockout is embryonic lethal, so conditional or knock-in humanized approaches would be required. This is a major gap in the modeling toolkit.

Limitations and Knowledge Gaps

1. **Evidence tier is low:** the literature is almost entirely case reports/series plus one 72-patient aggregated review. No prevalence/incidence, no controlled trials, no prospective natural-history cohorts.
2. **No genotype–phenotype correlation:** explicitly noted as lacking [PMID: 40120032]; it is unknown why some variants cause pediatric/rapidly progressive disease and others late-onset disease.
3. **Mechanistic heterogeneity unresolved:** different variants show opposite binding changes (integrin ↑ in one [PMID: 36774238] vs. endothelial/podocyte/heparin binding ↓ in another [PMID: 35774511]). A unifying biochemical model of how diverse binding defects all cause fibronectin deposition is not established.

4. **Source of circulating fibronectin not definitively localized:** allograft recurrence implicates a circulating (likely hepatic/plasma) mutant protein, but the tissue origin and whether lowering circulating mutant fibronectin would help are untested.
5. **No animal model:** absence of an in vivo GFND2 model limits preclinical therapeutic testing.
6. **No targeted therapy:** all management is nephroprotective/supportive; response to immunosuppression is inconsistent and likely reflects non-specific effects.
7. **Penetrance and expressivity not quantified.**

Proposed Follow-up Experiments / Actions

1. **Build a FN1 variant–function map:** systematically express the ~10+ catalogued pathogenic variants and measure binding to integrin, collagen IV, heparin, endothelial cells, and podocytes, plus fibril assembly, to test whether a common downstream assembly defect unifies the gain/alteration-of-function mechanism.
2. **Establish an in vivo model:** generate a knock-in mouse (e.g., Fn1 Y973C) or liver-specific conditional humanized FN1 model to overcome embryonic lethality and recapitulate glomerular deposition; evaluate whether circulating mutant fibronectin is necessary and sufficient.
3. **Test the circulating-protein hypothesis therapeutically:** in iPSC-derived kidney organoids and animal models, assess whether reducing circulating mutant fibronectin (e.g., ASO/siRNA knockdown of the mutant allele, or plasma clearance strategies) prevents/reverses deposition — directly relevant to allograft recurrence.
4. **Create a GFND2 registry and natural-history study** to derive prevalence, penetrance, genotype–phenotype correlations, and prognostic biomarkers.
5. **Standardize diagnostic workflow:** promote fibronectin IHC (IST-4/IST-9) + FN1 genetics + glomerular proteomics as a reflex pathway for MPGN/TMA-like biopsies of unknown cause, to reduce the frequent misdiagnosis.
6. **Transplant management protocol:** prospectively characterize allograft recurrence rates, timing, and whether pre/peri-transplant interventions (e.g., mutant-protein lowering) reduce recurrence.
7. **Explore sacubitril/valsartan and SGLT2 inhibitors prospectively** as nephroprotective agents given encouraging anecdotal reports.

Report compiled from 8 confirmed findings and 20 reviewed papers over 5 investigation iterations. All mechanistic and clinical claims are attributed to primary literature by PMID; direct quotes are drawn verbatim from the cited abstracts.